

Using Python to Predict Sales

Sales forecasting is very important to determine the inventory any business should keep. This article discusses a popular data set of the sales of video games to help analyse and predict sales efficiently. We will use this data to create visual representations. We won't dwell on the methodology and science behind statistics and demand forecasting, but will focus on understanding the basic steps needed to use Python and the Google Colab cloud environment to predict sales and arrive at an inventory strategy.



Several traditional tools like MS Excel have been used to derive predictive sales models, but there are lots of complexities, calculations, complicated formulae and cell manipulations that need to be used to do so. Instead, if we adopt an easy front-end application with a rich user experience, it can really help in making a better inference. So let's venture into the world of Python and its associated large set of statistical and analytical libraries to derive inferences without forcing the beneficiary to use complicated formulae, Excel manipulation, speculation, etc.

To derive this predictive model, we need to take a data set as an

example and understand the nature and relationships between its various attributes and forms. Figure 1 shows

a heuristic data set capturing the sales data of a company selling online games over the Internet and multiple

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
## Importing the dataset
dataset = pd.read_csv('../input/videogamesales/vgsales.csv')
```

Figure 1: Sample data set and use case from Kaggle (<https://www.kaggle.com/gregorut/videogamesales>)

